AN INVESTIGATION INTO OBESITY RATES

Researchers have been interested in studying the obesity trends worldwide over the past 25 years.

In 2014 the World Health Organisation (WHO) declared there were 1.6 billion overweight adults worldwide and 600 million of those are considered obese. Due to the concerns related to obese related diseases such as diabetes and heart diseases it has been of interest to see which countries are increasing and decreasing in obesity rates.

This task looks at the obesity rates in 1990 and 2010 for 26 randomly selected countries. The rates are percentages and are based on the total adult population for that country.

Is there a connection between a country's obesity rate in 1990 and 2010?

Is it possible to predict the trend of a country's obesity rate?

How has obesity changed in Australia over the past 16 years?

These questions and others are the focus of the task.

Using the data on the coloured paper to answer the following investigation questions.

Table 1: Obesity Rates

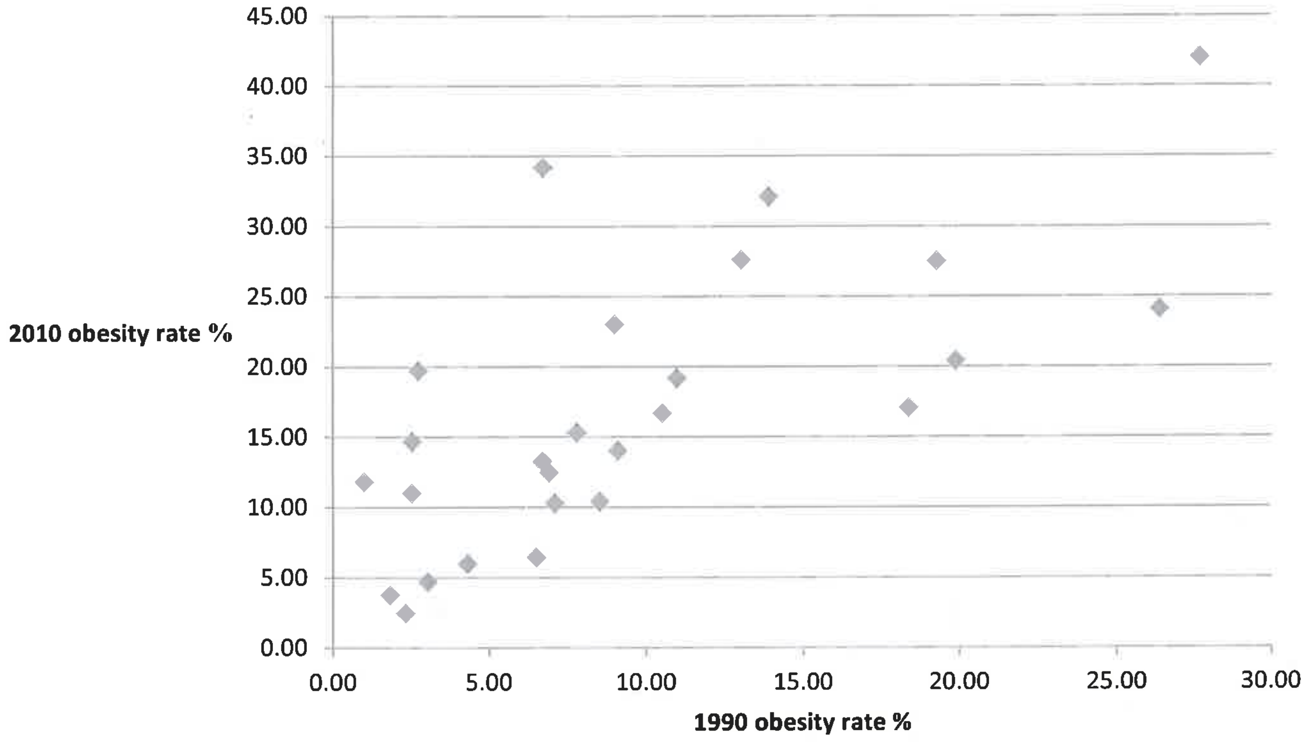
x y

	OBESITY RATE %	
Country	**1990**	**2010**
Australia	19.20	27.40
Austria	8.90	22.90
Belarus	18.30	17.20
Brazil	6.80	12.40
Canada	12.90	27.70
Chile	10.90	19.30
China	1.00	11.90
Congo	2.20	2.40
Cuba	2.60	19.60
Finland	19.80	20.50
India	2.40	10.90
Iran	2.40	14.60
Japan	1.70	3.70
Morocco	4.20	6.00
Netherlands	8.40	10.30
New Zealand	13.80	32.00
Nigeria	6.40	6.40
Oman	10.40	16.80
Russia	7.00	10.20
Saudi Arabia	26.30	24.00
Sweden	9.00	14.00
Thailand	2.90	4.60
Tunisia	6.60	13.20
Turkey	7.70	15.20
USA	27.60	41.90
Venezuela	6.60	34.10

Table 2: Average Obesity Rates

Year	Obesity Rate per 100,000
2000	5.3
2001	5.8
2002	6.3
2003	6.0
2004	6.7
2005	6.9
2006	6.6
2007	7.6
2008	7.3
2009	8.0
2010	8.7
2011	8.3
2012	9.0
2013	9.7
2014	10.7
2015	10.3
2016	11.2

Graph 1: Obesity Rates



Part One: Is there a trend between obesity rates from 1990 and 2010?

a) Calculate the five number summary statistics and IQR for the 1990 and 2010 obesity rates, correct to 1 decimal places.

- Put 1990 into list 1
Calc, one variable, ok
get sum stats

- Put 2010 into List 2
Calc, one variable
get sum stats

	1990	2010
Min	1.0	2.4
Q1	2.9	10.3
Med	7.4	14.9
Q3	12.9	22.9
Max	27.6	41.9
IQR	10.0	12.6

IQR → Q3 − Q1

* Make sure you put 1 decimal place for all.

(3 marks)
(O3)

b) Conduct an outlier test for the 1990 and 2010 obesity rates. Explain why/why not there is an outlier present, if one is present, state the value.

1990

Lower fence $Q1 - (1.5 \times IQR) =$
$2.9 - (1.5 \times 10) = -12.1$

Upper fence $Q3 + (1.5 \times IQR) =$
$12.9 + (1.5 \times 10) = 27.9$

No outliers for 1990 as all values inside fences of −12.1 → 27.9

2010

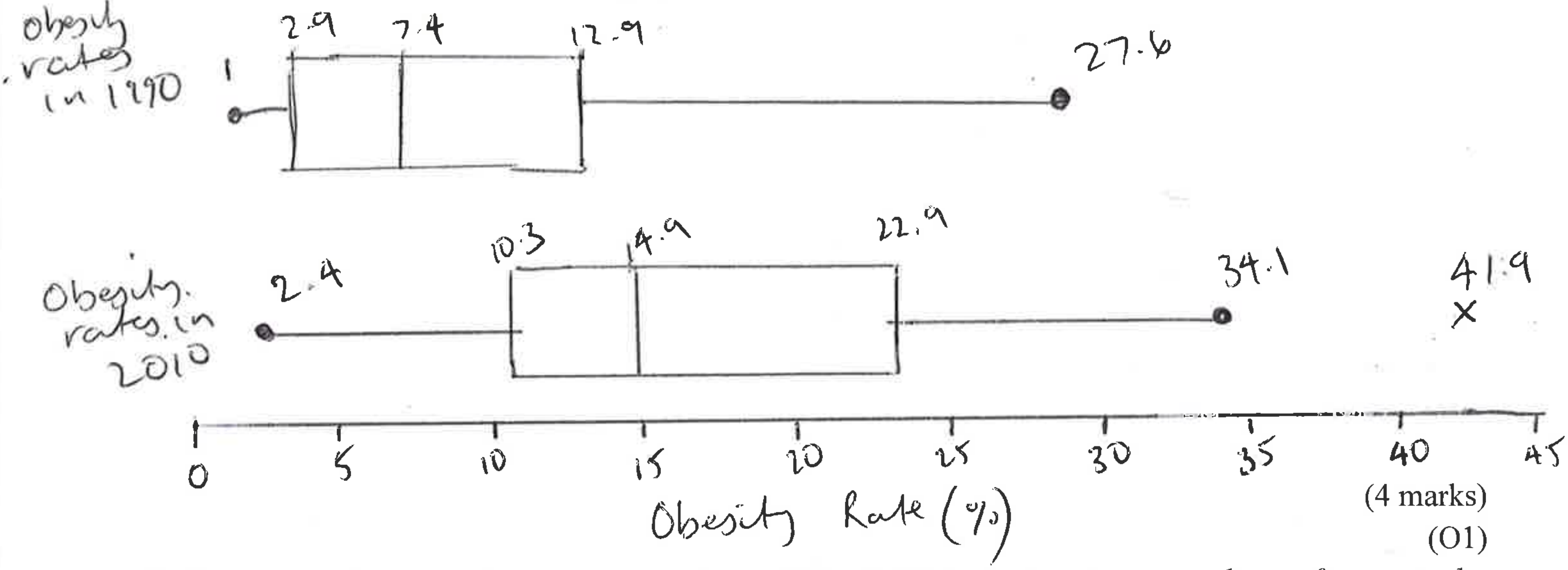
Lower fence $Q1 - (1.5 \times IQR)$
$10.3 - (1.5 \times 12.6) = -8.6$

Upper fence $Q3 + (1.5 \times IQR)$
$22.9 + (1.5 \times 12.6) = 41.8$

Yes, one upper outlier of 41.9 is present as it is above upper fence of 41.8

(4 marks)
(O1, O1, O2, O2)

c) Construct parallel boxplots on the axis below for the 1990 and 2010 Obesity rates (%). Label each boxplot clearly. Obesity rates in 1990 and 2010



(4 marks)
(O1)

d) Compare and contrast the obesity rates from 1990 and 2010. *Ensure you are making reference to the shape, centre, spread and quartiles of each year.*

Must say the following.

Obesity rates in 1990 and 2010 are both positively skewed. 2010 has the presence of 1 outlier of 41.9% and 1990 has no outliers -shape -outliers

Obesity rates in 2010 were generally higher ~~and~~ shown by the higher measure of centre with a median value of 14.9%, compared to 1990's ~~obes~~ median obesity rate of 7.4%. -centre.

2010 obesity rates were more spread with an IQR spanning over 12.6%. compared to 1990's IQR of 10%. - spread

The top 50% of obesity rates in 2010 was higher than Q3 in 1990 - quartile comment of your choice

(4 marks)
(O2)

e) Which measure of centre would be most appropriate to describe the central tendency of each data set (1990 and 2010)? Explain.

Due to the presence of an outlier and the skewness

(2 marks)
(O2)

Part Two: Comparing obesity rates in 2010.

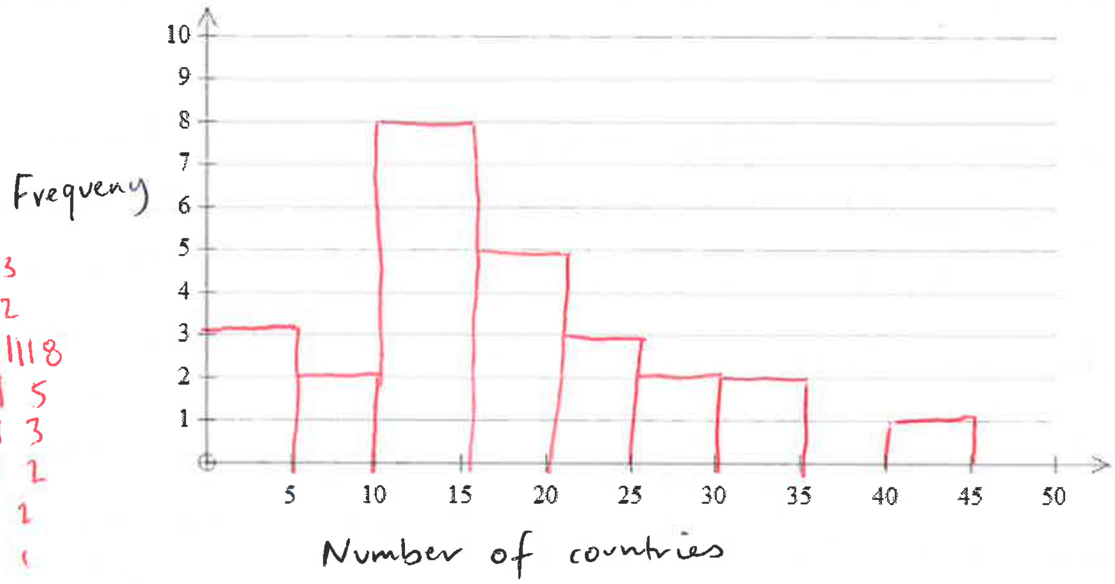
a) Determine the mean and standard deviation of the obesity rates in 2010 and place the values in the table below. Round all values to 1 decimal place.

	2010
Mean	16.9
Standard Deviation	9.9

(2 marks)
(O2)

b) Construct a histogram of the 2010 Obesity rate, with a width of 5. Label the scale on the axes below.

0-5 ||| 3
5-10 || 2
10-15 ~~||||~~ ||| 8
15-20 ~~||||~~ 5
20-25 ||| 3
25-30 || 2
30-35 || 2
35-40 0
40-45 | 1
45-50



(2 marks)
(O1, O3)

c) Describe the shape of the histogram.

Positively skewed with a possible outlier.

(1 mark)
(O1)

d) What percentage of countries have an obesity rate higher than 10% in 2010? State your answer to 2 decimal places.

26 - 5 below = 21

$\frac{21}{26} \times 100$ = 80.77% of countries have an obesity rate above 10% in 2010

(1 mark)
(O2)

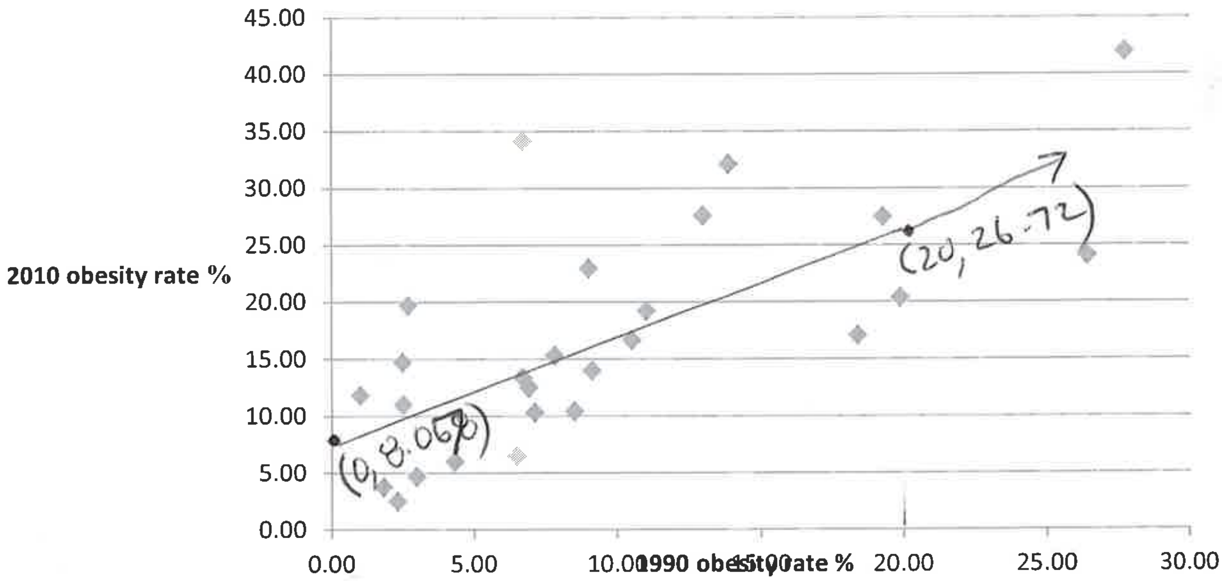
e) Why would it not be suitable to calculate the z-score for Australia in 2010?

Z (standardised score) can only be applied to symmetrical shaped data that fits a bell curve. This data is skewed.

(1 mark)
(O2)

Part Three: Regression Analysis

Below is a scatterplot showing the relationship between the obesity rates for 1990 and 2010 for the 26 countries listed in the data sample.



a) Write down the explanatory and response variables.

Explanatory = 1990 obesity rate
response = 2010 obesity rate.

(1 mark)
(O1)

b) Determine the equation of the least squares regression line for the scatterplot above. Write the equation in terms of the variables in the form $y = a + bx$ and state the coefficients correct to 3 significant figures. (Also write this equation in the space provided on your coloured data sheet)

Linear reg
$x \rightarrow$ list 1
$y \rightarrow$ list 2.

$y = 8.068 + 0.9322x$

to 3 sig figur

$y = 8.07 + 0.932x$

now put in variables.

2010 obesity rate = 8.07 + 0.932 × 1990 obesity rate.

(3 marks)
(O1, O2, O3)

c) Draw the least squares regression line found in part (b) on the scatterplot above. State the two points used to draw this line.

* Use y int of 8.07 to place.
* Trace another point from calc or substitute.
  when $x = 20, y = 26.72$.

(2 marks)
(O1, O2)

d) Interpret the slope of this least squares regression line in terms of the variables *1990 obesity rate* and *2010 obesity rate.*

$\frac{0.932}{1}$ As the 1990 obesity rate increases by 1%, the 2010 obesity rate increases by 0.932%.

(1 mark)
(O2)

e) Interpret the vertical intercept of the least squares regression line in terms of the variables *1990 obesity rate* and *2010 obesity rate*.

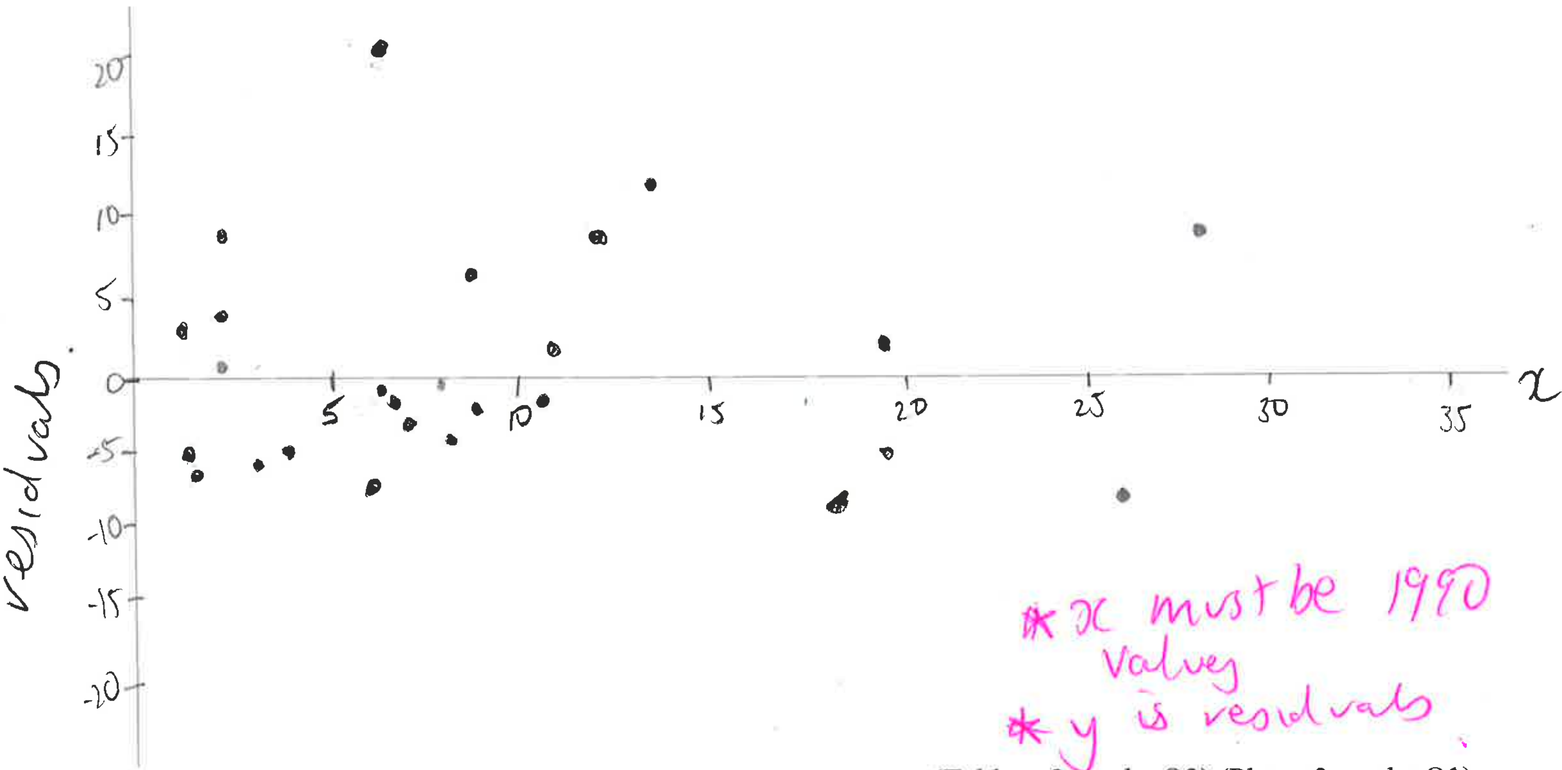
The y intercept means that ~~if~~ when the 1990 obesity rate is 0%, the 2010 rate will be 8.07 . .

(1 mark)
(O2)

f) Calculate the residual values to 1 decimal place in the following table for each country then plot on the axis provided on the next page.

1990 obesity rate	19.2	8.9	18.3	6.8	12.9	10.9	1	2.2	2.6	19.8	2.4	2.4	1.7
2010 actual obesity rate	27.4	22.9	17.2	12.4	27.7	19.3	11.9	2.4	19.6	20.5	10.9	14.6	3.7
2010 predicted obesity rate	26.0	16.4	25.1	14.4	20.1	18.2	9.0	10.1	10.5	26.5	10.3	10.5	9.7
Residual value	~~1.43~~ 1.4	6.5	-7.9	-2.0	7.6	1.1	2.9	-7.7	9.1	-6.0	0.6	4.3	-6.0

1990 obesity rate	4.2	8.4	13.8	6.4	10.4	7	26.3	9	2.9	6.6	7.7	27.6	6.6
2010 actual obesity rate	6.0	10.3	32.0	6.4	16.8	10.2	24.00	14.0	4.6	13.2	15.2	41.9	34.1
2010 predicted obesity rate	12.0	15.9	20.9	14.0	17.8	14.6	32.6	16.5	10.8	14.2	15.3	33.8	14.2
Residual value	-6.0	-5.6	11.1	-7.6	-1.0	-4.1	-8.6	-2.5	-6.2	-1.0	-0.0	8.1	19.9



* x must be 1990 values
* y is residuals

(Table = 3 marks O3) (Plot = 3 marks O1)

g) What does the residual analysis say about the original data?

The residual plot is random with no pattern, therefor the original data was linear.

(1 mark)
(O2)

h) Which country had the largest difference between the predicted 2010 and actual 2010 obesity rate?

Venezuela
(the largest residual)

(1 mark)
(O2)

i) Calculate the correlation coefficient (r) for the 26 countries correct to 3 decimal places and comment on what this indicates about the relationship between the prevalence of obesity in 1990 and in 2010.

0.697

This indicates a moderate positive linear relationship between 1990 and 2010 obesity rates.

(2 marks)
(O1, O3)

j) Calculate the coefficient of determination (r^2) for the 26 countries correct to 3 decimal places and comment on what this indicates about the relationship between the prevalence of obesity in 1990 and in 2010.

$r^2 = 0.486$

This indicates that 48.6% of the variation in 2010 obesity rates is due to the variation in ~~D~~ 1990 obesity rates.

(2 marks)
(O2, O3)

Part Four: Transforming the data

In an attempt to improve the linearity of the data, we will look at different transformations.

a) Suggest three possible transformations

y^2, $\log x$, $\frac{1}{x}$ or would aceept $\log(y)$ x^2 $\frac{1}{y}$

(1 marks)
(O2)

b) Apply each of the 3 transformation to your data. Complete the table below for the transformed data values. Use *y* to represent 2010 obesity rate and *x* to represent 1990 obesity rate in your equations. Round all values to 2 decimals places

Transformation	Equation of least squares regression line	Correlation Coefficient
y^2	$y^2 = 24.35 + 37.56x$	0.67
$\log x$	$y = 2.71 + 17.01 \log x$	0.66
$\frac{1}{x}$	$y = 21.61 - 21.79 \frac{1}{x}$	−0.48

(6 marks)

(O2, O2, O2, O3, O3, O3)

c) Comment on the improvement or otherwise of each transformation equation over the original equation you found in 3b. Use mathematical reasoning to support your comments, choose the best equation for calculating predicted values from the four equations considered (original from Q3b, plus 3 transformations).

None of the 3 transformations improved the value of r from the original equation. This means that the best least squares regression equation is the original equation.

(2 marks)
(O2)

d) Re-write your given equation, in terms of the variables given.

2010 obesity rate = 8.07 + 0.932 × 1990 obesity rate.

(1 mark)
(O2)

e) Use your chosen equation to predict a country's obesity rate in 2010 if it is 9.2% in 1990. Round your answer to the nearest whole number.

Substitute 9.2 for x.
y=16.64 The obesity rate would be 17%.

(1 mark)
(O2)

f) Use your chosen equation to predict a country's obesity rate in 2010 if it is 31% in 1990. Round your answer to the nearest whole number.

~~Solve~~ Substitute 31% for x
y=36.96 The obesity rate would be 37%.

(1 mark)
(O2)

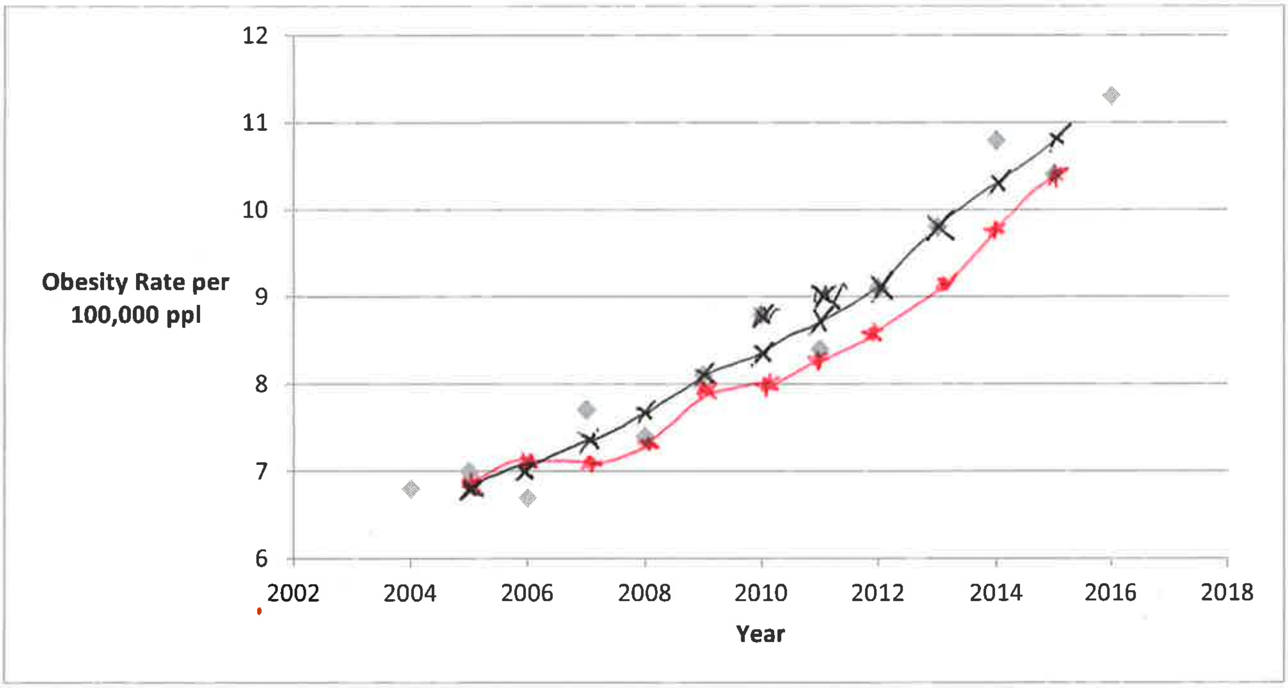
g) Which prediction from part e) and f) is more reliable. Explain

Making predictions for 2010 obesity rates are more reliable when using 9.2% for 1990 as it is within the data set (interpolation), compared to when making predictions using 31% for 1990 as that is extrapolation.

(2 marks)
(O2)

Part Five: Time Series

The time series graph below shows the obesity trend in Australia from 2000-2016 per 100,000 people.



a) Use this time series graph to describe the trend of the obesity rates in Australia from 2000 to 2016.

Secular Increasing random trend.

(1 mark)
(O1)

b) Smooth the graph above using a 3 point median smoothing graphical method. Mark in your points using an X in blue pen.

(2 marks)
(O2)

c) Use the table below to complete a 3 point moving mean. Round all values to 1 decimal place.

Year	Obesity Rate per 100,000 ppl	3 point moving mean
2004	6.3	
2005	7.5	6.9
2006	7	7.1
2007	6.8	7.1
2008	7.4	7.4
2009	8	8.0
2010	8. 5	8.0
2011	7. 5	8.3
2012	8.8	8.6
2013	9.4	9.1
2014	9.1	9.8
2015	10.8	10.4
2016	11.3	

(2 marks)
(O1, O3)

d) Plot the 3 point moving mean data onto the scatterplot above. Mark points as asterisks * using a red pen.

(2 marks)
(O1)

e) Comment on the effectiveness of the two smoothing techniques to identity any trends.

Both smoothing mean & smoothing median have worked to remove random fluctuations and highlight the increasing trend present.

(1 marks)
(O2)

Part Six: Displaying Data

1a Represent the data for Average obesity rates (from Table 2) in an ordered stem and leaf plot.

Key 5|3 = 5.3
= 530,000

5	3 8
6	0 3 6 7 9
7	3 6
8	0 3 7
9	0 7
10	3 7
11	2

(3 marks)
(O1)

g) What implications does this data have for the overall health of these nations with respect to time?

(1 mark)
(O2)

The obesity rates have ~~been~~ increased over time. This is shown from the county data in 1990 & 2010 and also from the average obesity rate table.
As obesity rates increase, the overall health of the nations may be decreasing.